Linear Algebra II

FMA_2F001_EP

Lecture Notes

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These are personal lecture notes and may contain errors. For the official course material consult the lecturer.

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1 Class 1: Basic Arithmetic

The arithmetic fact $1 + 1 = 2$ was shown by Russel and Whitehead. The proof operates using Peano's axioms for the natural numbers $\mathbb{N}$.

Theorem 1: Peano's Axioms for $\mathbb{N}$

- **Axiom 1:** $0 \in \mathbb{N}$.
- **Axiom 2:** There exists a successor function $S : \mathbb{N} \to \mathbb{N}$.

From these, we can define $1 = S(0)$ and $2 = S(S(0))$.

Definition 2: Addition

Addition is defined by the following rules:

1. $\forall m \in \mathbb{N} : m + 0 = m$ (Add1).
2. $\forall m, n \in \mathbb{N} : m + S(n) = S(m + n)$ (Add2).

Example 3: Computing $1 + 1$

$1 + 1 = 1 + S(0) \overset{\text{(Add2)}}{=} S(1 + 0) \overset{\text{(Add1)}}{=} S(1) = 2.$

Theorem 4: Peano's Axiom 9 (Induction)

Let P be a unary predicate. To prove $\forall m \in \mathbb{N}, P(m)$ is true, it is enough to prove:

- $P(0)$ is true.
- $\forall m : P(m) \implies P(S(m))$.

Theorem 5: Well Ordering Principle (WOP)

Every non-empty subset of $\mathbb{N}$ has a minimum.

Proposition 6: WOP implies Induction

The Well Ordering Principle can be used to prove the principle of Induction.

> Proof. Let $S = \{m \mid P(m) \text{ is not true}\}$. If $P(m)$ is not true for every m, then S is non-empty. We apply WOP to S, meaning there is a minimum element $m_0 = \min S$. But $m_0 \neq 0$, so $m_0 = S(m)$ for some m. This creates a contradiction. □

1.1 Relations on a Set

Definition 7: Relations and Equivalence

Let S be a set with elements $x, y \in S$. A relation on S is defined as a subset of $S \times S$. An equivalence relation must satisfy all three of the following properties:

- **Reflexivity:** $\forall x \in S, x \sim x$.
- **Symmetry:** $\forall x, y \in S, x \sim y \iff y \sim x$.
- **Transitivity:** $\forall x, y, z \in S, x \sim y$ and $y \sim z \implies x \sim z$.

Example 8: Equivalence Relations

"Same Parity" and "Same remainder mod n" are equivalence relations.

Remark 9: Partitions

An equivalence relation is the same as a partition of S.

Definition 10: Modular Classes

For every $n \in \mathbb{N}$, $a \equiv b \pmod{n}$ defines an equivalence relation. The set of equivalence classes for this relation is denoted $\mathbb{Z}/n\mathbb{Z}$.

1.2 Modular Arithmetic and Divisibility

Theorem 11: Chinese Remainder Theorem (Sunzi)

If m and n are coprime, there is a bijection between $\mathbb{Z}/mn\mathbb{Z}$ and $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$.

Example 12: Applying the Chinese Remainder Theorem

For $n = 2$ and $m = 3$, we can map between $\mathbb{Z}/6\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. If we want to find all $m \in \mathbb{Z}$ such that $m \equiv 2 \pmod{5}$ and $m \equiv 3 \pmod{7}$, the solution is the set $\{14 + 35k \mid k \in \mathbb{Z}\}$.

Theorem 13: Euclidean Division

$\forall a, \forall b \neq 0, \exists (q, r)$ such that $a = bq + r$ and $0 \leq r < |b|$.

Theorem 14: Bézout's Theorem

$\exists (u, v) \in \mathbb{Z}^2$ such that $au + bv = \gcd(a, b)$.

Example 15: Euclid's Algorithm

Applying the algorithm to find coefficients for 5 and 7:

- $7 = 1 \times 5 + 2$.
- $5 = 2 \times 2 + \gcd(5, 7)$.
- $1 = 5 - 2 \times 2 = 5 - 2 \times (7 - 5) = 3 \times 5 - 2 \times 7$.

We have found that $3 \times 5 \to (0, 1)$ and $-2 \times 7 \to (1, 0)$. If we want to find m such that $m \equiv 2 \pmod 5$ and $m \equiv 3 \pmod 7$, we compute $(3 \times 5) \times 3 + (-2 \times 7) \times 2$.

1.3 Prime Numbers

Definition 16: Prime Number

A prime number is a number $p \in \mathbb{N}$ such that $p \geq 2$ and p has only divisors 1 and p.

Theorem 17: Infinitude of Primes

There are infinitely many primes.

Theorem 18: Tao's Theorem

At least one of the following sets of primes (p, q) is infinite:

- **Twin primes:** $|p - q| = 2$.
- **Cousin primes:** $|p - q| = 4$.
- **Sexy primes:** $|p - q| = 6$.

Remark 19: Goldbach Conjecture

The Goldbach conjecture posits that every even number is equal to $p + q$ with p and q primes.

2 Lecture 2: Groups

Definition 1: Group

A group is a set G equipped with a binary operation (or group law) $*$, denoted $(G, *)$, that satisfies the following properties:

1. **Associativity:** $\forall a, b, c \in G, (a * b) * c = a * (b * c)$.
2. **Neutral Element:** $\exists e \in G$ such that $\forall a \in G, a * e = e * a = a$.
3. **Inverse Element:** $\forall a \in G, \exists b \in G$ such that $a * b = b * a = e$.

Optional: If a group also satisfies commutativity (i.e., $\forall a, b \in G, a * b = b * a$), it is called an **Abelian group**.

Example 2: Groups in Modular Arithmetic

- The set $\mathbb{Z}/n\mathbb{Z}$ with addition is a group. The neutral element is $[0]_n$ and the inverse of $[k]_n$ is $[-k]_n$.
- If p is prime, the non-zero elements modulo p under multiplication form a group. To find the inverse of k modulo p, one can use Bézout's identity on k and p. Because p is prime, there exist integers u and v such that $uk + vp = 1$. Modulo p, this simplifies to $uk \equiv 1 \pmod{p}$, which implies that u is the multiplicative inverse k^{-1}.

Definition 3: Powers and Cyclic Groups

For an element $g \in G$, its powers are defined as follows:

- For $n > 0$, $g^n = g * g * \cdots * g$ (n times).
- For $n = 0$, $g^0 = e$.
- For $n < 0$, $g^n = g^{-1} * g^{-1} * \cdots * g^{-1}$ ($|n|$ times).

A finite group G is called **cyclic** if it is generated by a single element g, meaning $G = \langle g \rangle = \{g^m \mid m \in \mathbb{Z}\}$.

Example 4: Common Groups

- **Dihedral groups** D_k: the group of symmetries of a regular k-gon.
- **Symmetric groups** Σ_n (or S_n): the group of permutations of n elements.
- **Alternating group** A_n: the subgroup of Σ_n containing all even permutations, i.e., $A_n = \{\sigma \in \Sigma_n \mid \varepsilon(\sigma) = 1\}$.
- The group of symmetries of polyhedra, such as the Rhombic Triacontahedron.

Definition 5: Subgroups and Morphisms

- A **subgroup** is a subset of G that forms a group under the same group law.
- A **morphism** (or homomorphism) is a map φ between groups that preserves the group structure, e.g., $\varphi(a * b) = \varphi(a) * \varphi(b)$.
- An **isomorphism** is a bijective group morphism.

Definition 6: Order of an Element

The order of an element $g \in G$ is the smallest positive integer m such that $g^m = e$. It is denoted $\omega(g) = \min\{m \geq 1 \mid g^m = e\}$.

Theorem 7: Weak Lagrange Theorem

Let G be a finite Abelian group, and let $g \in G$. Then the order of g divides the size of the group, i.e., $\omega(g) \mid \#G$.

PROOF. Let $\pi = \prod_{h \in G} h$ be the product of all elements in G. Consider the map $\varphi_g : G \to G$ defined by $h \mapsto g * h$. This map is a bijection, as it has an inverse $\varphi_{g^{-1}}$. Because G is Abelian, we can rearrange the product:

$$\pi = \prod_{h \in G} h = \prod_{h \in G} \varphi_g(h) = \prod_{h \in G} (g * h)$$

Factoring out g for each of the $\#G$ elements, we get $\pi = g^{\#G} \cdot \prod_{h \in G} h = g^{\#G} \cdot \pi$. By canceling π, we find $g^{\#G} = e$.
We can write $\#G = q \cdot \omega(g) + r$ using Euclidean division, with $0 \leq r < \omega(g)$. Substituting this into our equation gives:

$$e = g^{\#G} = g^{q\omega(g)+r} = (g^{\omega(g)})^q \cdot g^r = e^q \cdot g^r = g^r$$

Since $g^r = e$ and $0 \leq r < \omega(g)$, the minimality of the order implies $r = 0$. Therefore, $\omega(g)$ divides $\#G$. □

Theorem 8: Strong Lagrange Theorem

If G is a finite group and $H \subset G$ is a subgroup, then $\#H$ divides $\#G$.

PROOF. We say two elements $x, y \in G$ are equivalent modulo H ($x \sim y$) if $y^{-1}x \in H$. This condition is equivalent to $x \in yH$, where yH represents a left coset. The left cosets form a partition of the group G, so $G = \bigsqcup_{y_i \in I} y_i H$ for some index set I of representatives.
Every coset $y_i H$ has exactly the same size, which is equal to $\#H$. Thus, we can conclude that the total number of elements in G is given by $\#G = \#H \times \#I$. This immediately shows that $\#H \mid \#G$. □

Theorem 9: Cauchy's Theorem

Let G be a finite group and let p be a prime number such that $p \mid \#G$. Then there exists an element $g \in G$ such that $\omega(g) = p$.

PROOF. Consider the set $S_p = \{(g_1, g_2, \ldots, g_p) \mid g_i \in G, g_1 g_2 \ldots g_p = e\}$. We define an equivalence relation on S_p using cyclic shifts: two tuples are equivalent if one is a cyclic permutation of the other (i.e., shifting the sequence by some offset k). Notice that the size of S_p is $(\#G)^{p-1}$, because the first $p-1$ elements can be chosen freely, and the final element g_p is uniquely determined as $(g_1 \ldots g_{p-1})^{-1}$.
Under this equivalence relation, every equivalence class contains either 1 or p elements. The tuple $(e, \ldots, e)$ forms an equivalence class of size 1. We can partition S_p into these classes, yielding the equation $\#S_p = N_1 + pN_p$, where N_1 is the number of classes of size 1, and N_p is the number of classes of size p.
Since $p \mid \#G$, we know $p \mid (\#G)^{p-1}$, so $p \mid \#S_p$. Thus, $p \mid (N_1 + pN_p)$, which implies $p \mid N_1$. We already know $N_1 \geq 1$ because of the class containing $(e, \ldots, e)$. Because $p \mid N_1$, we must have $N_1 \geq p \geq 2$.
This means there is at least one other class of size 1, which corresponds to a tuple $(g, g, \ldots, g) \in S_p$ for some $g \neq e$. Since this tuple is in S_p, it must satisfy $g^p = e$. Because p is prime and $g \neq e$, the order of g is exactly p, so $\omega(g) = p$. □

3 Lecture 3: Rings

Definition 1: Unital Ring

A unital ring is a set R equipped with two binary operations, denoted $(R, +, \times)$, satisfying the following properties:

1. $(R, +)$ is an Abelian group, with a neutral element denoted 0 or 0_R.
2. Multiplication "$\times$" is associative: $\forall x, y, z \in R, x(yz) = (xy)z$.
3. Multiplication "$\times$" has a neutral element, denoted 1 or 1_R.
4. Distributivity of multiplication over addition: $\forall a, b, c \in R, a(b + c) = ab + ac$ and $(b + c)a = ba + ca$.

Note: $(R, \times)$ and $(R\backslash\{0\}, \times)$ are not necessarily groups, because elements do not generally have multiplicative inverses.
If the multiplication operation is also commutative, then R is called a **commutative ring**.

Definition 2: Ring Morphism

A map $\varphi : (R_1, +, \times) \to (R_2, +, \times)$ is a ring morphism if:

1. $\varphi : (R_1, +) \to (R_2, +)$ is a group morphism.
2. $\varphi(1_{R_1}) = 1_{R_2}$.
3. $\forall a, b \in R_1, \varphi(ab) = \varphi(a)\varphi(b)$.

Theorem 3: The Ring of Integers as an Initial Object

The ring $(\mathbb{Z}, +, \times)$ is the paradigmatic example of a ring, and it serves as the initial object in the category of unital rings. This means that for any ring R, there exists a unique ring morphism $\psi : \mathbb{Z} \to R$. Therefore, $\mathbb{Z}$ is the unique initial object up to ring isomorphism.

Proof. By "abstract nonsense", let R_0 be another initial object. Then there exist unique morphisms $\psi_1 : \mathbb{Z} \to R_0$ and $\psi_2 : R_0 \to \mathbb{Z}$. The composition $\psi_2 \circ \psi_1 : \mathbb{Z} \to \mathbb{Z}$ must be the identity morphism (since $\mathbb{Z}$ is initial), which implies $\psi_1 = \psi_2^{-1}$, establishing a ring isomorphism between $\mathbb{Z}$ and R_0. □

3.1 Subrings and Ideals

Definition 4: Subring

A subring is a subset that forms a ring under the same (induced) algebraic structure. For example, we have the chain of subrings: $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C} \subset \mathbb{H}$.

Remark 5: Ideals vs. Subrings

For groups, the kernel of a group morphism is a subgroup. However, for rings, the kernel is generally *not* a subring (it usually does not contain 1_R). The correct notion to capture

the behavior of kernels in rings is an **ideal**.

Definition 6: Ideal

An ideal $I \subset R$ is a subgroup of $(R, +)$ that is stable under multiplication by any element of R. Equivalently: $\forall a, b \in R$ and $\forall x, y \in I$, we have $ax + by \in I$ (analogous to a linear combination in a vector space).

Proposition 7: Kernels are Ideals

Let $\varphi : R_1 \to R_2$ be a ring morphism. Then $\ker(\varphi) = \{x \in R_1 \mid \varphi(x) = 0\}$ is an ideal of R_1 (but not a subring!).

Proof. Let $a, b \in R_1$ and $x, y \in \ker(\varphi)$. We want to prove that $ax + by \in \ker(\varphi)$, which means $\varphi(ax + by) = 0$. Using the properties of ring morphisms:

$$\varphi(ax + by) = \varphi(ax) + \varphi(by) = \varphi(a)\varphi(x) + \varphi(b)\varphi(y)$$

Since $x, y \in \ker(\varphi)$, we know $\varphi(x) = 0$ and $\varphi(y) = 0$. Thus, $\varphi(a) \cdot 0 + \varphi(b) \cdot 0 = 0$. □

Definition 8: Principal Ideal

In a commutative ring R, ideals of the form $(x) = xR = \{xy \mid y \in R\}$ are called **principal ideals**. For example, all ideals of $\mathbb{Z}$ are principal (of the form $m\mathbb{Z}$).

3.2 Quotient Rings

Theorem 9: Quotient Ring Construction

For any commutative ring R and any ideal $I \subset R$, we can construct a quotient space R/I by defining an equivalence relation $r_1 \sim r_2 \iff r_1 - r_2 \in I$. The equivalence classes (or cosets) are denoted $[r] = r + I$.
The set of these classes, R/I, inherits a natural ring structure given by:

- $[r_1] + [r_2] = [r_1 + r_2]$
- $[r_1] \times [r_2] = [r_1 \times r_2]$

Example 10: Modular Arithmetic and CRT

Let $R = \mathbb{Z}$ and $I = m\mathbb{Z}$. This construction gives the usual quotient ring $\mathbb{Z}/m\mathbb{Z}$. If m and n are coprime, the Chinese Remainder Theorem establishes a ring isomorphism: $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$.

3.3 Taxonomy of Rings

Definition 11: Types of Rings

- **Integral Domain:** A commutative ring where $xy = 0 \implies x = 0$ or $y = 0$ (i.e., it has no zero divisors).
- **Principal Ideal Domain (PID):** An integral domain where every ideal is a principal ideal.
- **Division Ring:** A ring where $(R \setminus \{0\}, \times)$ is a group (i.e., every non-zero element has a multiplicative inverse).
- **Field:** A commutative division ring (i.e., $(R \setminus \{0\}, \times)$ is an Abelian group).

Remark 12: Examples and Hierarchy

We have the following strict inclusions for commutative rings:

$$\text{Commutative Rings} \supset \text{Integral Domains} \supset \text{PIDs} \supset \text{Fields}$$

- **Non-commutative Rings:** $M_2(\mathbb{R}), M_3(\mathbb{R})$.
- **Division Rings (not fields):** The quaternions $\mathbb{H}$.
- **Fields:** $\mathbb{R}, \mathbb{Q}, \mathbb{Z}/p\mathbb{Z}$ (where p is prime).

3.4 Special Ideals

Definition 13: Prime and Maximal Ideals

Let $I \subsetneq R$ be a proper ideal in a commutative ring R.

- I is a **prime ideal** if $xy \in I \implies x \in I$ or $y \in I$. For example, $2\mathbb{Z}$ is a prime ideal, but $4\mathbb{Z}$ is not.
- I is a **maximal ideal** if for any ideal J such that $I \subseteq J \subsetneq R$, it must be that $I = J$.

Theorem 14: Quotients by Prime and Maximal Ideals

Let R be a commutative ring and I be a proper ideal.

- I is a maximal ideal $\iff$ $(R/I, +, \times)$ is a field.
- I is a prime ideal $\iff$ $(R/I, +, \times)$ is an integral domain.

Because every field is an integral domain, this immediately implies that every maximal ideal is also a prime ideal.